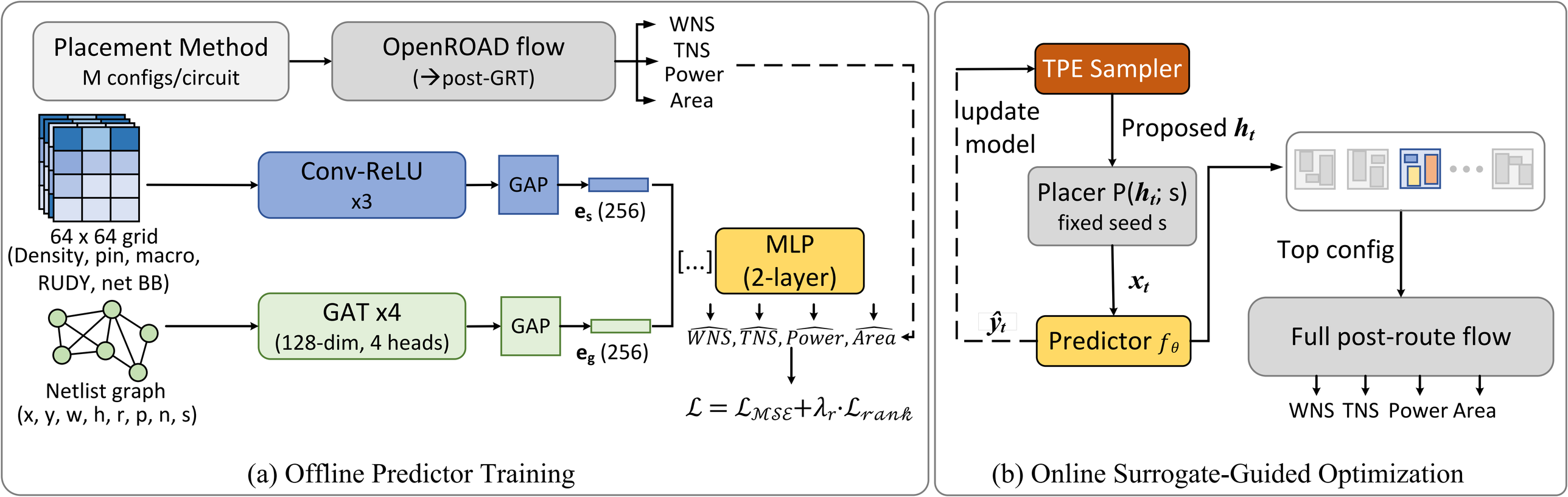
Placement Method
M configs/circuit
OpenROAD flow
(→post-GRT)
WNS
TNS
Power
Area
64 x 64 grid
(Density, pin, macro,
RUDY, net BB)
Conv-ReLU
x3
GAP
$\mathbf{e}_s$ (256)
[...]
MLP
(2-layer)
Netlist graph
(x, y, w, h, r, p, n, s)
GAT x4
(128-dim, 4 heads)
GAP
$\mathbf{e}_g$ (256)
$\widehat{WNS}, \widehat{TNS}, \widehat{Power}, \widehat{Area}$
$\mathcal{L} = \mathcal{L}_{MSE} + \lambda_r \cdot \mathcal{L}_{rank}$
(a) Offline Predictor Training
TPE Sampler
update
model
Proposed $\boldsymbol{h}_t$
Placer P($\boldsymbol{h}_t$; s)
fixed seed s
$\boldsymbol{x}_t$
$\hat{\boldsymbol{y}}_t$
Predictor f_θ
Top config
Full post-route flow
WNS TNS Power Area
(b) Online Surrogate-Guided Optimization